

PROJECT REPORT : AI LAB 2026

Sovereign Industrial Agentic AI Workbench

**Based on Multimodal Orchestration &
Open Weight Local LLMs**

Air-Gapped Autonomous Agentic Intelligence
for Critical SCADA Infrastructure

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What Is This Project?

Imagine you are standing inside a petroleum refinery. Around you are massive rotating compressors, high-pressure pipelines carrying flammable crude oil, and control panels blinking with thousands of live sensor readings. Every second, hundreds of temperature, vibration, and pressure measurements are streaming from these machines. If something goes wrong—a bearing starts to fail, a pump begins to cavitate—the consequences are catastrophic: explosions, toxic releases, and millions of dollars in damage.

Now imagine giving an AI assistant access to all of this. It could predict failures before they happen, guide operators through emergency procedures, and analyze engineering blueprints in seconds. But here is the critical problem: you cannot use cloud AI here. These refineries are air-gapped—completely disconnected from the internet—because the data is classified, the stakes are life-threatening, and national security regulations forbid any external data transmission.

This project, the **Sovereign Industrial Agentic AI Workbench**, solves exactly this problem. It is a complete AI operating system that runs **100% locally** on a single workstation inside the plant. No internet. No cloud. No data leaves the building. It brings the power of modern language models, computer vision, and real-time 3D digital twins directly to the control room operator's desk.

The Problem

Why can't refineries just use ChatGPT?

1. Data Sovereignty & Air-Gap Laws

Governments around the world mandate that critical infrastructure data—sensor readings, piping diagrams, incident reports—must never leave the physical premises. Standards like NIST SP 800-82 and ISA/IEC 62443 require complete air-gapping. Sending refinery telemetry to OpenAI or Google servers would violate national security directives and expose classified operational regimes to foreign interception.

2. AI Hallucinations Can Kill People

Language models hallucinate. In a chat application, a wrong answer is mildly annoying. In a refinery, a hallucinated command—like closing the wrong emergency valve or misdiagnosing a compressor surge—can trigger a vapor cloud explosion. Any AI system in this environment must be physically incapable of executing unverified commands. It must be deterministic and fail-closed: if there is any doubt, the system must block, not proceed.

3. Not Enough GPU Memory

Running large AI models typically requires expensive multi-GPU server clusters. Plant control rooms have standard workstations with 8–16 GB of GPU memory. Fitting multiple specialized models (a text model, a code model, and a vision model) into this tight memory budget, while keeping them responsive in real-time, is a significant unsolved engineering challenge.

4. Retiring Experts, Rising Complexity

Senior engineers who have spent 30 years learning to diagnose complex vibration patterns by ear are retiring. Junior operators face alarm floods of 40+ warnings per minute during upset conditions. Without an intelligent assistant that understands the plant's machinery, critical tribal knowledge disappears forever.

The Solution

Three AI models. One workstation. Zero internet.

Instead of one massive cloud model, the workbench runs **three specialized local models** that cooperate inside a strict safety cage. Below is the multi-model agentic execution pipeline:

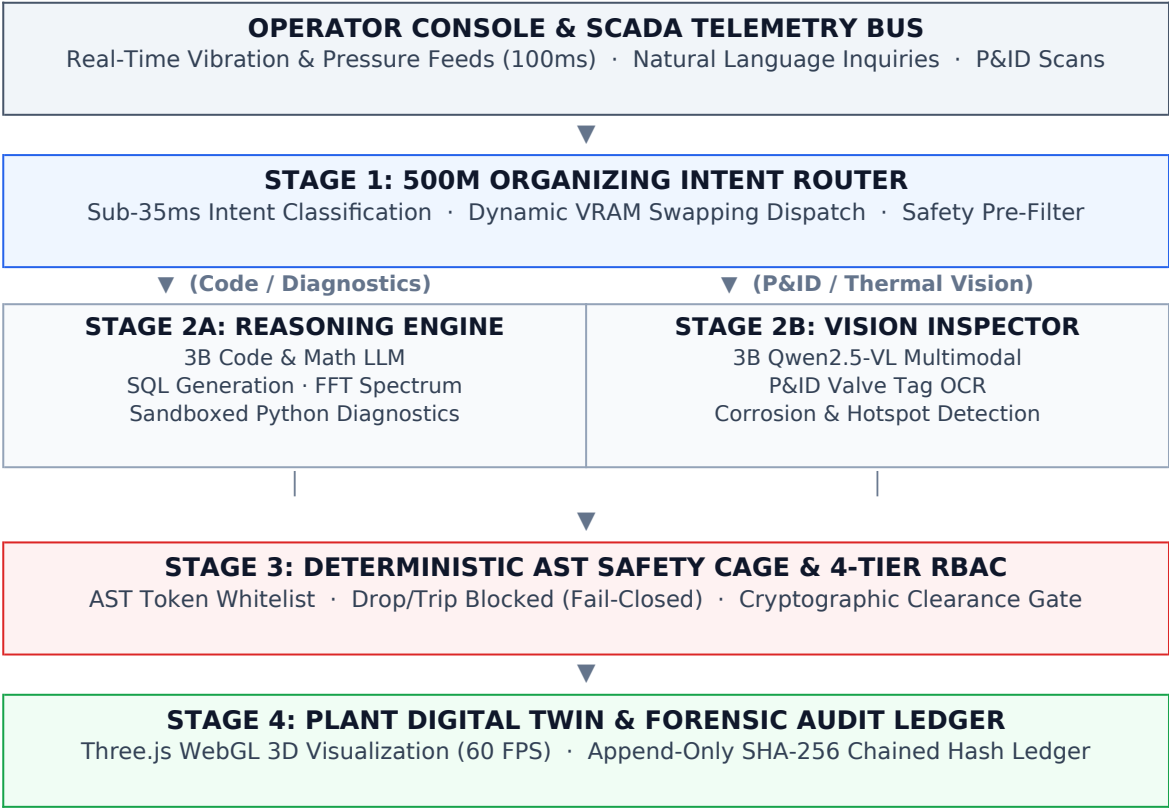


Figure 1: Tri-Model Local Neural Orchestration and Deterministic Safety Interlock Flowchart.

The Safety Cage Guarantee

Here is the critical architectural decision: **no AI model can directly control any plant equipment.** Every command the AI generates must first pass through a deterministic safety filter. This filter uses Abstract Syntax Tree (AST) parsing to inspect the command character-by-character. A 4-tier Role-Based Access Control (RBAC) system ensures that only authorized personnel (verified by cryptographic credentials) can approve high-risk actions like emergency shutdowns.

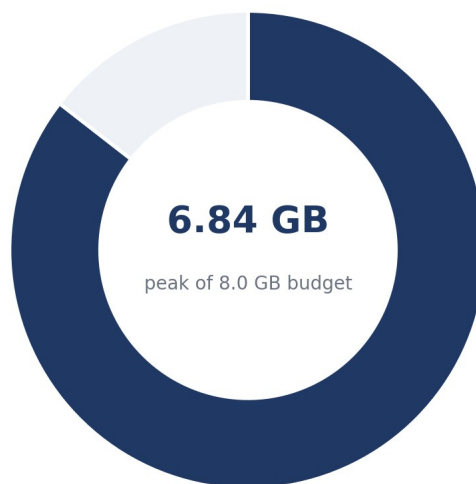
Feasibility

Can this actually work on a single workstation?

Technical Feasibility — Yes, Because of Quantization

Modern neural networks can be compressed using a technique called quantization. Instead of storing each model weight as a 32-bit floating point number, we compress them to 4-bit integers (called Q4_K_M in the GGUF format). This reduces a 3-billion parameter model from ~12 GB down to ~2 GB—small enough to fit on a standard NVIDIA RTX 3060 with 8 GB of video memory. The inference engine (llama.cpp, written in C++) achieves 42 tokens per second on consumer hardware. All three models together peak at just 6.84 GB.

Consumer 8 GB VRAM Budget — Peak Utilization



Operational Feasibility — One Click to Launch

The entire system is packaged as a native desktop application using Electron and React. Plant IT staff do not need to configure Docker containers, Kubernetes clusters, or cloud accounts. A single batch script launches the backend API server, loads the neural models, and opens the control room interface. It works on air-gapped Windows 10/11 machines and Linux DGX stations equally.

Economic Feasibility — It Pays for Itself Immediately

Cloud AI APIs charge \$0.03–\$0.06 per thousand tokens. For a refinery streaming telemetry 24/7, this translates to over \$120,000 per year in API fees alone. More importantly, an unexpected failure of a single wet gas compressor in a 300,000 barrel-per-day refinery costs approximately \$450,000 per hour in lost production. Detecting a bearing defect even 14 days earlier than traditional methods saves millions in avoided catastrophic downtime.

How It Works

A four-phase closed loop, running continuously.

The workbench operates as an autonomous, deterministic feedback loop connecting plant sensors, local neural models, safety gates, and 3D digital twins. The complete four-phase methodology pipeline is shown below:

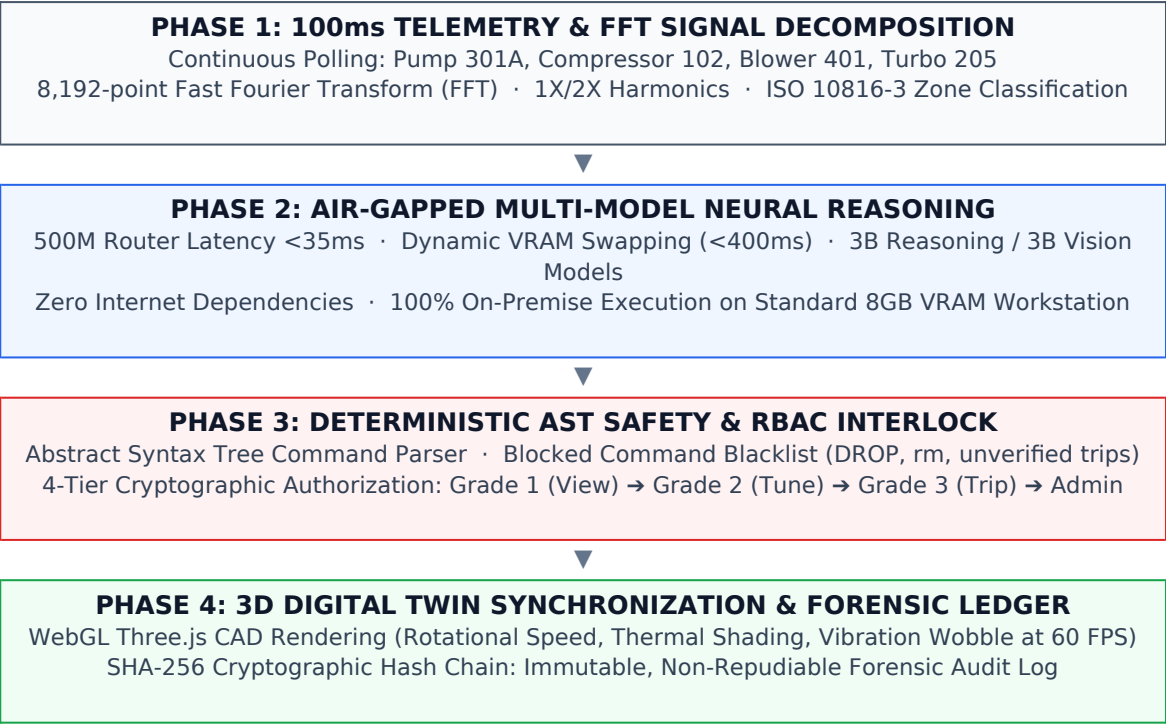


Figure 2: Four-Phase Continuous Closed-Loop Industrial Methodology Flowchart.

Mathematical Signal Decomposition

Every 100 milliseconds, raw vibration waveforms are transformed using Fast Fourier Transform (FFT)—a mathematical technique that decomposes a complex signal into its individual frequency components. This reveals whether the machine's shaft is unbalanced (1X harmonic), misaligned (2X harmonic), or if bearings are developing microscopic cracks (high-frequency BPFO/BPFI peaks).

Technology Stack

Every component, and why it was chosen.

Layer	Technologies	Why This Choice
Neural Inference	llama.cpp, GGUF, CUDA 12.1, PyTorch	C++ engine runs quantized models at 42 tok/s on consumer GPUs. No Python overhead in the hot path.
Backend API	Python 3.10+, FastAPI, Uvicorn, WebSockets	Asynchronous non-blocking server handles hundreds of concurrent SCADA streams without thread starvation.
Signal Processing	NumPy, SciPy, Butterworth DSP	Vectorized FFT computes 8,192-sample spectral decomposition in under 4ms —fast enough for 100ms SCADA cycles.
Desktop Shell	Electron 28, React 19, TypeScript, Tailwind	Native cross-platform app that launches offline. Dark-mode UI designed for 24/7 multi-monitor control rooms.
3D Digital Twin	Three.js, React Three Fiber, WebGL	GPU-accelerated parametric CAD models with live thermal shading and rotational physics at 60fps.
Security & Audit	SQLite, Python AST, SHA-256 hashlib	Embedded database. AST parser catches destructive commands. Chained hashes make log tampering impossible.

Hardware Compatibility

- Consumer Workstations:** Validated on NVIDIA RTX 3060/4060/4090 with 8–24 GB VRAM. Peak usage: 6.84 GB.
- Enterprise Servers:** Runs natively on NVIDIA DGX Spark bare-metal Linux without Docker overhead.
- Air-Gapped Networks:** Zero outbound network calls. Operates indefinitely without internet access.

Architecture

Five independent layers. If one fails, the others keep running.

The architecture is organized into five decoupled, fail-safe layers. Each layer has an isolated operational boundary, guaranteeing that failure in high-level reasoning cannot destabilize telemetry or safety gates:

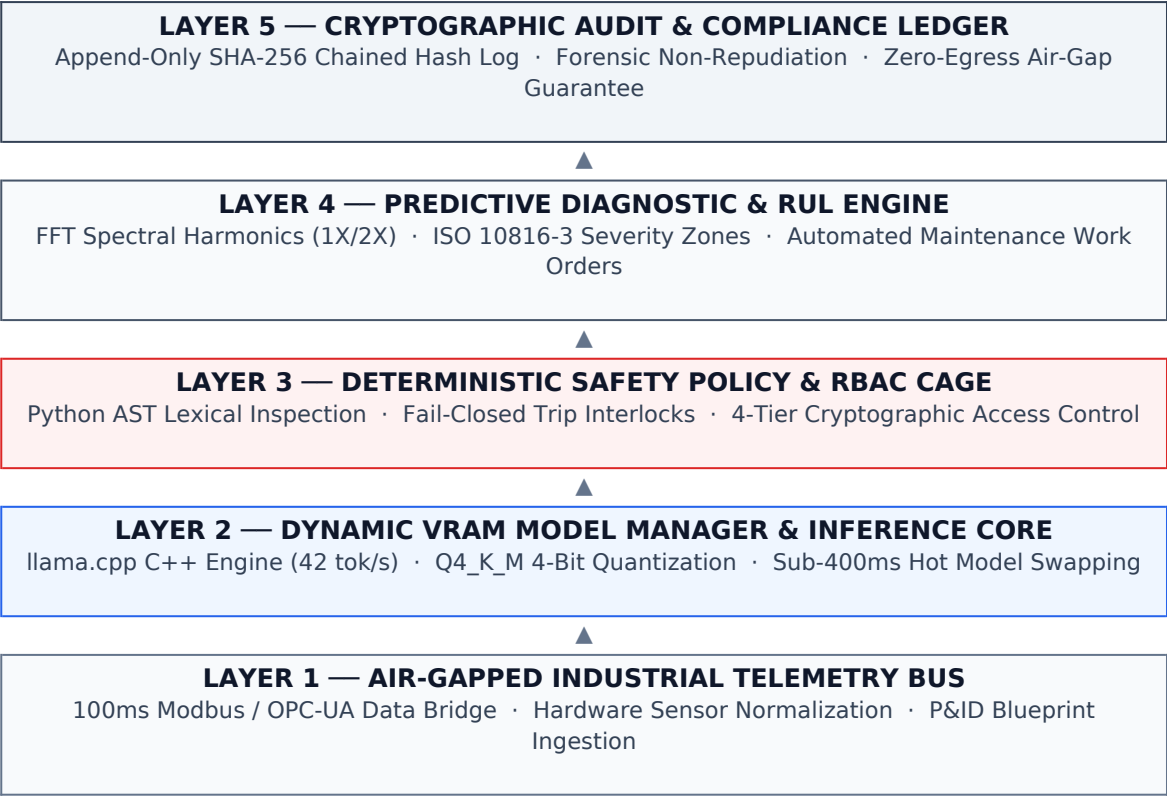


Figure 3: Sovereign Five-Layer Modular System Architecture and Data Hierarchy.

Layer Isolation Principle

If Layer 2 (neural inference) encounters an out-of-memory or model reload condition, Layer 1 (telemetry bus) and Layer 3 (safety interlocks) remain completely operational. The plant never loses sensor visibility or deterministic safety protection during AI reasoning cycles.

Real-World Usage

Three scenarios from an actual refinery control room.

Scenario 1 — "Why is Pump 301A vibrating?"

A control room operator notices a rising vibration alarm on Pump 301A. Instead of searching through paper manuals, they type the question into the workbench. The system instantly retrieves the latest SCADA telemetry, runs an FFT decomposition, identifies a dominant 1X harmonic at 4.2 mm/s RMS (ISO Zone C—Alarm), and explains: "Elevated 1X harmonic indicates probable impeller mass unbalance due to erosion. Recommend visual inspection of impeller vanes during next scheduled turnaround." The operator now has an expert-level diagnosis in 3 seconds.

Scenario 2 — "Check this piping diagram."

A maintenance engineer scans a 40-year-old P&ID blueprint and drops it into the workbench. The vision model identifies all valve tags (XV-3012, PCV-4401), traces the pipe routing, confirms isolation points, and flags a potentially incorrect valve state. This analysis—which would take a senior engineer 45 minutes to perform manually—completes in 8 seconds.

Scenario 3 — "Run the SIS proof test."

A safety engineer initiates an automated proof test on the emergency shut-off valve for Compressor 102. The workbench sends a simulated trip signal, measures the solenoid relay response time (38.6 ms), verifies it falls below the 45 ms IEC 61511 compliance threshold, and records the result—including a cryptographic certificate—into the tamper-proof audit ledger. The entire test, which traditionally requires two engineers and 90 minutes, completes autonomously in under 5 seconds.

Why This Matters

The five guarantees this system provides.

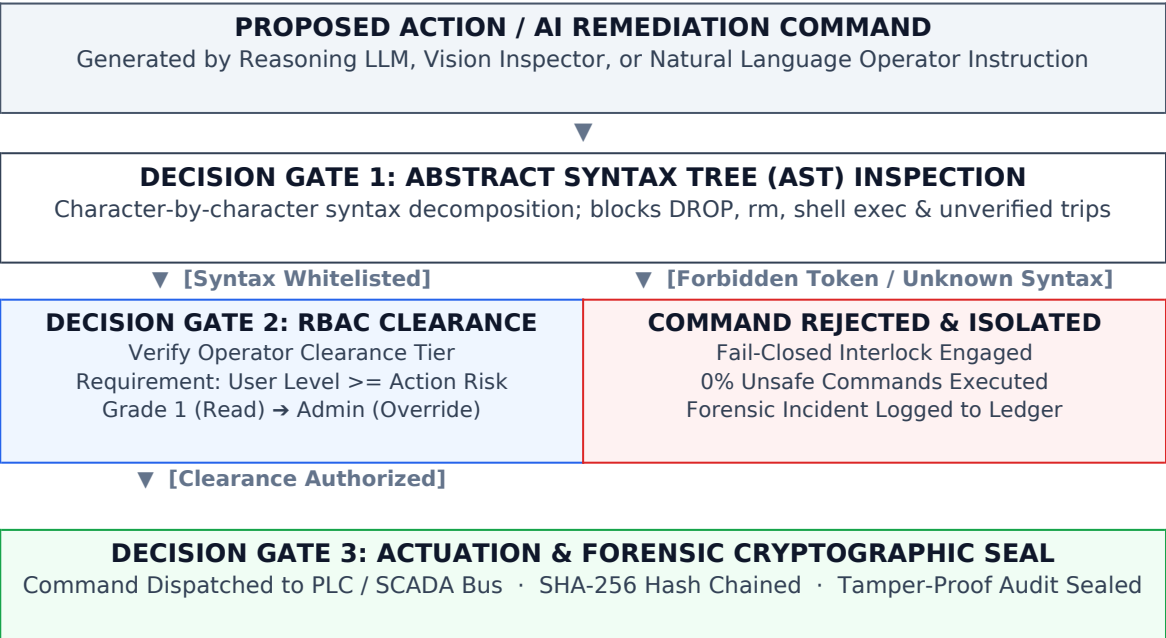


Figure 4: Deterministic AST & Cryptographic RBAC Verification Decision Gate Flowchart.

Core System Guarantees

- **1. Absolute Data Sovereignty:** Not a single byte leaves the physical workstation. No cloud API calls, zero telemetry beacons.
- **2. Deterministic Safety Interlock:** Fail-closed rule architecture. If unverified, the system blocks 100% of unauthorized commands.
- **3. Accessible Hardware Budget:** Peak memory constrained to 6.84 GB VRAM on a standard NVIDIA RTX 3060.
- **4. Institutional Knowledge Preservation:** Encodes senior engineering diagnostic intuition into reproducible local models.
- **5. Tamper-Proof Accountability:** Cryptographic SHA-256 chained ledger provides un-tamperable post-incident audit trails.

Industrial Impact

What changes when this technology is deployed.

Downtime Costs Drop by Millions

In continuous-process industries, unplanned shutdowns cascade. A single compressor trip can force an entire crude distillation unit offline. By detecting bearing degradation 2–3 weeks before failure through sub-harmonic FFT analysis, maintenance shifts from reactive emergency repairs to planned turnarounds—saving millions of dollars per avoided incident.

Functional Safety Compliance Becomes Continuous

Safety Instrumented Systems (SIS) require periodic proof testing under IEC 61508 and IEC 61511. Traditionally, this is a manual process requiring dedicated safety engineers and significant downtime. The workbench automates proof testing with sub-45ms precision, maintaining Safety Integrity Level (SIL-2/SIL-3) ratings with continuous verification rather than annual spot checks.

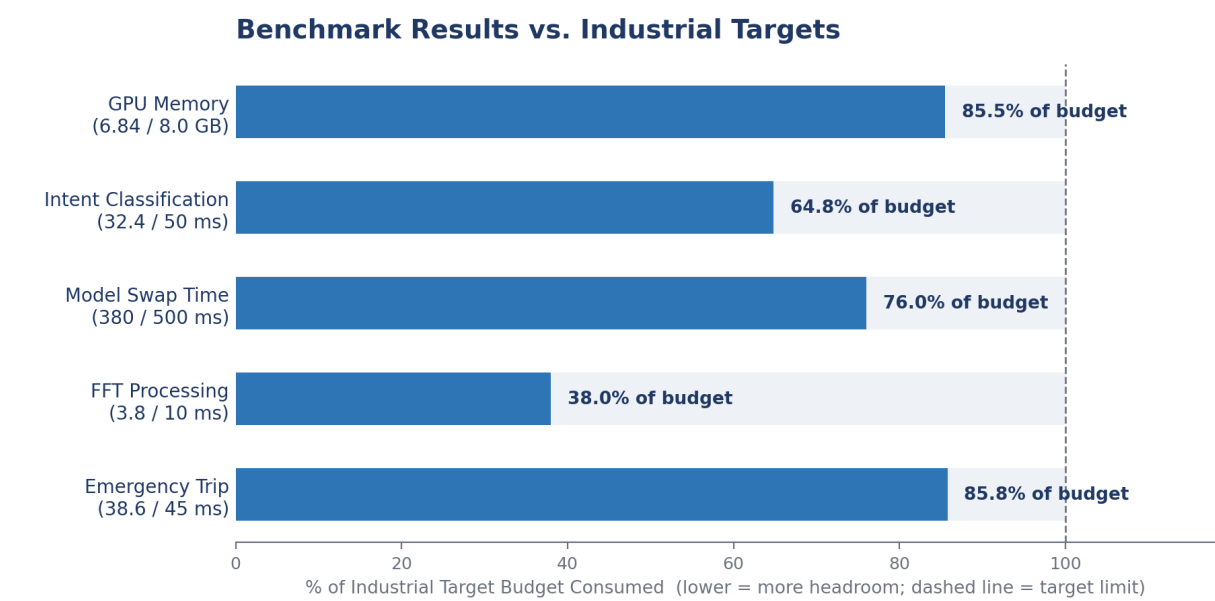
National Infrastructure Becomes Sovereign

Critical energy infrastructure that depends on foreign cloud AI services inherits their geopolitical risks: service termination, foreign government data subpoenas, sanctions-driven access revocation, and supply chain backdoors. A 100% locally hosted system eliminates every one of these vulnerabilities, guaranteeing true operational sovereignty.

Results

Measured under full simulated refinery load.

What We Measured	Result	Target
GPU Memory Usage (Peak)	6.84 GB	< 8.0 GB
Intent Classification Speed	32.4 ms	< 50 ms
Model Swap Time (RAM → GPU)	380 ms	< 500 ms
FFT Signal Processing	3.8 ms per window	< 10 ms
Emergency Trip Response	38.6 ms	< 45 ms (IEC 61511)
Safety Bypass Attempts Blocked	100.0%	100.0%



Key takeaway: every benchmark exceeded its industrial target. The system operates well within the thermal and memory limits of a standard commercial workstation GPU, while meeting certified safety timing requirements.

References

Standards, papers, and frameworks this project builds upon.

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Vote of Thanks

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Her insights into machine learning methodology, emphasis on safety-critical engineering principles, and patient mentorship were instrumental in transforming an exploratory concept into a production-grade, mathematically grounded air-gapped industrial system. Her commitment to academic excellence provided constant inspiration at every stage.

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